BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Sound and Compression

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How does a bitmap represent pixel colours?

2. Why can heavy virtual-memory use slow a program?

A) RAM becomes permanent storage
B) The CPU loses every core
C) All instructions are deleted
D) Secondary storage is slower

3. "A parameter is the final printed output of every program." Correct this. Explain why.

4. A microphone records digital sound. "Sampling stores every instant of the sound wave." Correct this. How does it affect recorded detail?

Today's objectives

Recall the essential ideas of sound and compression. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Sampling
Measuring an analogue signal at regular time intervals.

Sample rate
Number of samples recorded each second.

Bit depth
Number of bits per sample.

Lossless compression
Reduces size without losing original information.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Analogue to digital	An analogue sound wave is sampled at regular time intervals, and each sample is quantised to an available binary level.
Sample rate	Sample rate counts samples taken per second. More samples can capture changes in the wave more accurately.
Bit depth and size	Bit depth controls possible amplitude levels. Uncompressed mono size in bits = sample rate x duration x bit depth.
Compression	Lossless methods reconstruct original data exactly; lossy methods discard some information to achieve smaller files.

I Do - One worked example

Calculate the uncompressed size of a 10-second mono recording sampled at 8000 Hz with 8-bit depth. [4 marks]

1. Identify three factors

2. 8000 samples/second, 10 seconds, 8 bits/sample.

3. Total samples = 8000 x 10 = 80,000.

4. Size = 80,000 x 8 = 640,000 bits = 80,000 bytes. [4]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Calculate the uncompressed size of 5 seconds of mono audio at 16,000 samples/s and 8 bits/sample. [4]

2. Explain the effect of increasing sampling rate. [4]

3. Compare lossy and lossless compression for an audio archive. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the number of bits in a mono uncompressed recording.

```
rate = 800
seconds = 2
bitDepth = 8
bits = ...
print(bits)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. A mono recording lasts 4 seconds, with 4000 samples/second and 8 bits/sample. Calculate the uncompressed size in bytes. [4]

Q3. Explain one effect of increasing bit depth while sample rate and duration stay the same. [3]

Q4. Explain why lossless compression is more suitable than lossy compression for an original master recording. [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table can help check how values change during a program. ☐ T ☐ F
2. An analogue sound wave is sampled at regular time intervals, and each sample is quantised to an availabl... ☐ T ☐ F
3. Sample rate counts samples taken per second ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Each pixel has a colour code stored in binary. ☐

Q2 D - Secondary storage is slower ☐

Q3 A named input received by a subprogram when called with an argument. ☐

Q4 Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured. ☐

Code First - Model

```
rate = 800
seconds = 2
bitDepth = 8
bits = rate * seconds * bitDepth
print(bits)
```

Output: 12800

We Do - Check the model

1. 16,000 x 5=80,000 samples [1]; x 8=640,000 bits [2]; / 8=80,000 bytes [1]. ☐

2. More samples each second [1] can capture the original sound wave more accurately [1], but increases uncompressed data stored per second [1] and therefore file size [1]. ☐

3. Lossless preserves all original information [1] and can restore identical sound data [1]; lossy removes some information [1] for potentially greater size reduction but cannot restore exact original [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 4000 x 4=16,000 samples [1]; x 8=128,000 bits [2]; / 8=16,000 bytes [1]. ☐

Q3 More amplitude levels [1] can improve quantisation precision [1], while increasing uncompressed file size [1]. ☐

Q4 Lossless preserves all original data [1], so the original can be reconstructed exactly [1], avoiding irreversible quality loss [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: